SIGMA

Scaled Impact & Growth Metric Algorithm

Production-Grade Ranking System | Bayesian · IRT · Elo Hybrid | Full Mathematical Reference

Overview

SIGMA estimates latent user capability — not mere activity. It fuses Item Response Theory (IRT), Bayesian inference, and Elo-style comparative updating into one framework, augmented by super-exponential repeat decay, Shannon entropy diversity enforcement, velocity-based anti-farming, and a logistically compressed display scale.

Component	Model	Role in SIGMA
IRT 3PL	Item Response Theory	Task difficulty vs. user capability
Bayesian	Laplace approx.	Uncertainty-aware rating updates
Elo	Dynamic K-factor	Peer comparison & relative ranking
Decay	Super-exponential	Anti-farming, repeat suppression
Entropy	Shannon H	Diversity enforcement
Logistic	Compressed sigmoid	Non-linear tier progression

Input Variables per Task

Variable	Symbol	Range	Description
Failure Rate	F	[0, 1]	Fraction of attempts that fail
Risk Level	R	[1, 5]	Risk on a 5-point scale
Rarity	r	(0, 1]	Fraction of population who attempted
Impact Score	I	[1, 10]	Real-world significance
Verification Status	V	0/0.3/0.8	0 verified, 0.3 partial, 0.8 none
Repeat Count	n	{1,2,3,...}	Times this task repeated by user
Time Required	T	hours	Calibrates rarity and failure rate

Equation 1 — Task Difficulty Score D(t)

```
D(t) = b1*ln(1+F) + b2*R^2 + b3*ln(1+1/r) + b4*I^y + b5*(1 - V_penalty)
```

- b1=2.1 b2=1.8 b3=1.5 b4=1.2 b5=3.0 y(gamma)=1.4
- ln(1+F) -- logarithm stabilises as F approaches 1
- R^2 -- risk squared, high-risk tasks exponentially harder
- ln(1+1/r) -- rare tasks score much higher; r capped at 0.001
- I^y -- super-linear impact: impact 10 is 2.64x impact 5, not 2x
- V_penalty -- unverified tasks lose 80% of verification contribution

Equation 2 — Repeat Decay V_effective(n)

```
V_effective(n) = D(t) * exp( -0.35 * (n-1)^2 )
```

- n=1 -> 1.000 (full value)
- n=2 -> 0.705 (70.5%)
- n=3 -> 0.247 (24.7%)
- n=4 -> 0.043 (4.3%)
- n=5 -> 0.004 (0.4%)

Note: The (n-1)^2 exponent means decay accelerates with each repetition — farming is killed.

Equation 3 — IRT Probability of Success P(success)

```
P(success | O, a, b, c) = c + (1-c) * [ 1 / (1 + exp(-a*(O-b))) ]
```

- O (theta) = user capability (latent, estimated)
- a = task discrimination parameter in (0, 3]
- b = task difficulty on logit scale = (D(t) - mean_D) / std_D
- c = guessing/floor parameter in [0, 0.25]
- O >> b => P approaches 1 (easy for this user)
- O << b => P approaches c (hard; c is the floor, never zero)

Equation 4 — Bayesian Capability Update

```
Prior: O ~ N(u0, s0^2)
Likelihood: L(x|O) = P^x * (1-P)^(1-x)
u_new = u_old + 0.08 * (x - P(O_old)) * b_normalized
s^2_new = s^2_old * (1 - 0.15 * P * (1-P))
```

- eta=0.08 (learning rate) kappa=0.15 (confidence tightening)
- Succeeding on a much-harder-than-expected task gives a large positive update
- Every outcome shrinks s^2 -- system grows more certain about user over time

Equation 5 — Elo Comparative Update

```
E_A = 1 / (1 + 10^((O_B - O_A) / 400))
```

```
Delta_O_A = K * D(t) * (S_A - E_A)
```

- E_A = expected probability A outperforms B on this task
- S_A = actual outcome (1 = completed, 0 = failed)
- D(t) scales the update -- harder tasks move ratings more

Equation 6 — Dynamic K-Factor K(theta, n)

```
K(O, n) = 32 / (1 + 0.05*O + 0.80*ln(1+n))
```

- New user O=0, n=0 -> K = 32.00
- Intermediate O=2, n=100 -> K = 6.67
- Elite O=4, n=500 -> K = 5.21

Note: Elite users cannot spike their rating from a single task. New users move fast.

Equation 7 — Combined SIGMA Rating Update

```
O_new = O_old
+ 0.50 * Delta_O_Bayesian
+ 0.30 * Delta_O_Elo
+ 0.15 * C(u) (consistency bonus)
- 0.05 * Psi(t) (inactivity decay)
```

Equation 8 — Consistency Bonus C(u)

```
C(u) = 0.50 * (1 - exp(-0.05 * streak_days)) * (CV + 0.10)^(-1)
```

- streak_days = consecutive days with at least one verified completion
- CV = std(task_difficulties) / mean(task_difficulties) over last 30 days
- Low CV (consistent effort) -> denominator near 1 -> full bonus
- Erratic bursting -> high CV -> bonus suppressed

Equation 9 — Inactivity Decay Psi(t)

```
O(t) = O_floor + (O_current - O_floor) * exp(-0.003 * Delta_t)
O_floor = 0.30 * O_peak
```

- At Delta_t = 90 days: retain 76% above floor
- At Delta_t = 365 days: retain 33% above floor
- You never decay below 30% of your all-time peak

Equation 10 — Anti-Gaming: Velocity Cap

```
z = (tasks_per_day - pop_mean) / pop_std
penalty = 1 / (1 + exp(z - 3.0))
```

- At z=3.0 -> penalty = 0.50 (gains halved)
- At z=5.0 -> penalty = 0.12 (88% suppression)
- Smooth logistic form -- no hard cutoff that can be gamed at the boundary

Equation 11 — Anti-Gaming: Entropy Diversity Check H(u)

```
H(u) = - Sum_i [ p_i * log2(p_i) ]
diversity_multiplier = min(1.0, H(u) / log2(N_categories))
```

- p_i = fraction of completed tasks in category i
- Only one category used -> H=0 -> multiplier=0 -> near-zero gains
- All categories equal -> H=H_max -> multiplier=1.0 -> full gains

Equation 12 — Final SIGMA Score (Display)

```
SIGMA = 1000 * [ 1 / (1 + exp(-1.20 * (O - 2.50))) ]
```

Note: Moving 800->850 requires ~3x the raw theta gain as moving 400->450. Every extra point at the top costs exponentially more.

Tier Thresholds

Tier	Label	SIGMA Score	Percentile
1	Novice	< 300	Bottom 40%
2	Skilled	300 - 499	40th - 65th
3	Expert	500 - 699	65th - 85th
4	Elite	700 - 849	85th - 95th
5	Master	850 - 949	95th - 99th
6	Apex	>= 950	Top 1%

Equation 13 — Worked Example: Funded Startup Launch

Task: launch a VC-funded startup (first attempt, fully verified, public record).

Variable	Value	Interpretation
F	0.90	90% of people fail to get funded
R	5	Maximum risk level
r	0.02	2% of population has attempted this
I	9	Very high real-world impact
V	0.0	Fully verified (public funding record)
n	1	First time -- no repeat decay applied

Step 1 -- Compute D(t)

```
D(t) = 2.1*ln(1.90) + 1.8*25 + 1.5*ln(51) + 1.2*9^1.4 + 3.0*1.0
= 1.35 + 45.00 + 5.90 + 23.72 + 3.00
= 78.97
```

Step 2 -- Repeat decay (n=1)

```
V_effective = 78.97 * exp(0) = 78.97
```

Step 3 -- Second funded startup (n=2)

```
V_effective = 78.97 * exp(-0.35) = 78.97 * 0.705 = 55.67
```

Step 4 -- Bayesian update (user theta=2.0, task b=3.5)

```
P = 0.1 + 0.9 * [1 / (1 + exp(-1.5*(2.0-3.5)))]
= 0.1 + 0.9 * 0.095 = 0.186 (18.6% expected success rate)
Delta_O = 0.08 * (1 - 0.186) * 1.0 = +0.065
```

Large positive update: user succeeded at something with only 18.6% expected probability.

Pseudocode

```
function process_task_completion(user, task, outcome):
D = b1*ln(1+F) + b2*R^2 + b3*ln(1+1/r) + b4*I^y + b5*(1-V_penalty)
V = D * exp(-0.35 * (repeat_count - 1)^2)
b = (D - global_mean_D) / global_std_D
P = c + (1-c) / (1 + exp(-a * (user.theta - b)))
delta_bayes = 0.08 * (outcome - P) * b
user.sigma = user.sigma * (1 - 0.15 * P * (1-P))
user.theta += delta_bayes
E_user = 1 / (1 + 10^((theta_peer - user.theta) / 400))
K = 32 / (1 + 0.05*user.theta + 0.80*ln(1+user.n_tasks))
delta_elo = K * V_norm * (outcome - E_user)
z = (user.tasks_today - pop_mean) / pop_std
velocity_pen = 1 / (1 + exp(z - 3.0))
H = -sum(p_i * log2(p_i) for each category)
div_mult = min(1.0, H / log2(N_categories))
CV = std(recent_difficulties) / mean(recent_difficulties)
C = 0.50 * (1 - exp(-0.05*streak_days)) / (CV + 0.10)
eff_elo = delta_elo * velocity_pen * div_mult
user.theta += 0.50*delta_bayes + 0.30*eff_elo + 0.15*C
user.theta = theta_floor + (user.theta - theta_floor)
* exp(-0.003 * days_since_last_task)
SIGMA = 1000 / (1 + exp(-1.2 * (user.theta - 2.5)))
user.n_tasks += 1
return SIGMA
```

Master Parameter Table

Symbol	Value	Meaning
b1 - b5	2.1, 1.8, 1.5, 1.2, 3.0	Difficulty weight vector
y (gamma)	1.4	Impact super-linearity exponent
l (lambda)	0.35	Repeat decay steepness
h (eta)	0.08	Bayesian learning rate
k (kappa)	0.15	Confidence tightening factor
K_max	32	Max Elo update magnitude
a, b (K)	0.05, 0.80	K-factor dampening terms
w1 - w4	0.50, 0.30, 0.15, 0.05	Combined update weights
d (delta)	0.05	Streak growth rate

p (psi)	0.003	Inactivity decay constant (half-life 231 days)
z_threshold	3.0	Velocity anomaly trigger (sigma above mean)
a_scale	1.20	Final score logistic steepness
O_midpoint	2.50	Theta value mapping to SIGMA score 500

All parameters are tunable hyperparameters. Re-estimate quarterly via maximum likelihood on held-out user-task outcomes using an EM algorithm.

Known Weaknesses and Exploits

Weakness	Description	Mitigation
Verification oracle	Fake/colluded verification collapses the system	Blockchain timestamps, institutional sign-off, third-party audit
Cold start	New users have high uncertainty; early tasks overshadow	Mandatory 10-task calibration battery at onboarding
Penalises specialists	Neurosurgeons score low diversity	Profession declaration sets floor diversity_multiplier = 0.6
Laplace skew	Posteriors skewed at extreme theta values	Particle filters or variational inference for top/bottom 1%
Global mean Elo	Fast growth inflates if matched to global mean	Stratify peer groups by theta range, not global mean
Parameter drift	Beta weights stale as task demographics shift	Quarterly EM re-estimation on held-out outcome data